

Gerardo J. Luna

Mechanical Engineer | Manufacturing Engineer

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Objective

Motivated engineering student with over 5 years of experience in manufacturing processes, seeking a Space Suit Technician position. Skilled in precision fabrication, equipment operation, and laboratory procedures, with experience supporting research in ISRU materials and thermal systems for aerospace applications.

Education

Cullen College of Engineering, University of Houston, Houston, Texas

B.Sc. in Mechanical Engineering, May 2027

LSC Cy-Fair, Lone Star College, Houston, Texas

A.S. in Mechanical Engineering, May 2026

A.A.S. in Mechatronics, Robotics, and Automation Engineering Technology, December 2021

Skills

Software & Programming: C++, Python, Microsoft Excel, NX Siemens, SolidWorks, Fusion 360, LabVIEW, PSPICE

Lab Equipment: Thinky Planetary Mixer, Thermal Vacuum Oven, Taber Abrasion Testing

Manufacturing Equipment: CNC Machining, Welding, Soldering, Additive Manufacturing

Work Experience

HOME DEPOT, HOUSTON, TX 2018-Present

Merchandise Execution Associate

- Co-led coordination with strategic partners and major suppliers to implement merchandising and installation projects, including constructing fixtures and display apertures according to Standardized Planograms (SPOG) to ensure accurate layout and safety.
- Coordinated with store leadership to support weekly maintenance, safety checks, and pricing updates to ensure previous projects complied with store policies and met safety codes.

Academic Research & Projects

NASA L'Space NPWEE Summer 2025: Project Varco Winning Team 2025-2026

Mechanical Engineer/ Materials Research Scientist

- Co-led an 11-person team developing vacuum-sprayed ceramic coating to mitigate dust abrasion for Lunar infrastructure and inflatable habitats. In a massive team effort, the following deliverables were submitted: Technology Readiness Plan (TRP), New Technology Report (NTR), and a 10-page winning proposal. The project is currently entering Phase two, in which we will partner with NASA to deliver our final ceramic-sprayed substrates and performance test results.
- Developed 10 Standard Operating Procedures and Project Trackers for Manufacturing and Coupon Testing, using the following lab and manufacturing equipment: Thermal Vacuum Oven, LRD grit blasting, Thinky Mixers, and Non-destructive material characterization tools across 3 partner laboratories.

Leadership

American Society of Mechanical Engineers (ASME)

Vice-Chair

- Coordinated monthly chapter meetings, workshops, and two conference events connecting students with industry recruiters. Managed a maker space supporting students' research projects and hands-on fabrication resources.
- Led two NASA Student Competitions (NASA Lunabotics and NASA Gateway to Blue Skies), overseeing the system design, budgets, and technical documentation to deliver a university-level research experience at a community college.